The Idea of a Base — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1	**(b) powers of n.** Each landmark is the previous one × n.
Q2	**(b) base-10.** Its landmark numbers are 1, 10, 100, 1000…
Q3	**1033.** 143 = 1×125 + 0×25 + 3×5 + 3×1.
Q4	**(a) 143.** 1×125 + 0×25 + 3×5 + 3.
Q5	**True.** The base tells you how many of one landmark make the next.
Q6	**(a) Both true, R explains A.** 100 = 4×25 + 0×5 + 0 = 400 in base 5.

Short answer

Q7	**2244 (base 5).** 324 = 2×125 + 2×25 + 4×5 + 4.
Q8	**288.** 2×125 + 1×25 + 2×5 + 3 = 250 + 25 + 10 + 3.
Q9	**1, 5, 25, 125, 625, 3125.** (Powers of 5.)

Long answer

Q10	**(i) 400 in base 5 (ii) 1210 in base 4.** 100 = 4×25 (base 5); 100 = 1×64 + 2×16 + 1×4 + 0 (base 4).
Q11	**16 = 100 in base 4.** 16 is exactly the landmark number 4^2 = 16, so it is 1×16 + 0×4 + 0×1.
Q12	**Because landmark numbers are powers, their product is another landmark number** (e.g. $10^2 \times 10^3 = 10^5$). The Roman system has irregular landmark sizes, so multiplication there is much harder.

Total: 21 marks.